The Q-Foundational Stack

A Complete Computing Architecture from Number Representation to Artificial Intelligence

Registry: [?]

Series Path: [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [?] → [?]

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Domain: Machine Learning / Integer Arithmetic / Neural Network Training

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Classification: Theory of Everything from First Principles

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Abstract

We present the Q-Foundational Stack: a unified computing architecture in which a single representational atom — the VFR triple (Value, Factor, Remainder) — serves as the basis for number representation, instruction encoding, processor design, memory architecture, operating system, application framework, 3D rendering, network security, and artificial intelligence. The stack eliminates floating-point arithmetic, cache coherence, branch prediction, speculative execution, shared memory, runtime exceptions, and the distinction between code and data. In their place: exact integer arithmetic, isolated core memory, deterministic execution, and a universal batch processing model. The result is a system that runs 100,000 game entities at 60 frames per second on a $343 development board consuming 5 watts, with security guaranteed by physical topology rather than software policy, and with inline neural-symbolic intelligence that cannot hallucinate by construction. Ten specifications define the complete stack from transistor to artificial mind. Every

component uses the same data format, the same instruction set, and the same processing model. The complexity of modern computing is not intrinsic to computation. It is intrinsic to decisions made in the 1980s about how to represent numbers and organize memory. Different decisions produce a different world.

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1. The Atom

1.1 VFR: Value, Factor, Remainder

Every quantity in the Q-Foundational Stack is represented as:

[V: i32, R: i8] 5 bytes

V = value (the dividend)

R = remainder (of V divided by the batch factor F)

F = declared once per batch, shared across all values

Division is the only arithmetic operation that escapes integer space. Every other operation on integers produces an integer. Floating-point exists because of division. VFR does not evaluate the division. It preserves it. `7 / 3` is not `2.333...` — it is `[7, 3, 1]`. The quotient is recoverable. The remainder is exact. No information is destroyed.

The base is 32. Division by the base is a right shift by 5 bits. The remainder is a bitwise AND with `0x1F`. Both are single-cycle integer operations. The most common operation in the system — positional decomposition — costs one shift and one mask.

1.2 Scale Through Powers of 32

The factor F selects a power of 32 as the unit scale. The base unit (F=1) is smaller than the Planck length. F=65 exceeds the observable universe.

F=1: 32^-1 sub-Planck (base unit)

F=2: 32^0 = 1 Planck length

F=37: 32^35 human heart

F=40: 32^38 particles in a human body

F=65: 32^63 observable universe

The entire range of physical reality spans 64 steps in F. That fits in a 7-bit integer. The value V stays small at every scale. The remainder R provides sub-unit precision in an i8.

There is no underflow because the base unit is below the Planck scale — physics itself cannot go lower. There is no overflow because F=65 exceeds everything that exists. There is no NaN because every integer operation produces a valid integer. There is no infinity because the representable range already exceeds the physical universe.

1.3 What VFR Eliminates

Every mechanism in IEEE 754 floating-point exists to manage the consequences of approximation. VFR does not approximate. The mechanisms become unnecessary:

- Epsilon comparison: eliminated. Values are exact. `a == b` is a single integer compare.
- NaN propagation: eliminated. No operation produces an invalid result.
- Denormalized numbers: eliminated. No normalization exists.
- Rounding modes: eliminated. No rounding occurs.
- Warp divergence: eliminated. No epsilon branches means all parallel lanes take the same path.
- Non-determinism: eliminated. Integer arithmetic gives the same result on every platform, every run.

These are not features removed. They are consequences of a decision never made.

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2. The Container

2.1 The Batch

Every dataset in the stack is a batch: a sequence of VFR pairs preceded by a 7-byte header.

[F: i16] [count: u32] [depth: u8]

[V, R] [V, R] [V, R] …

F declares the domain and scale. Count declares the quantity. Depth declares the shape — how many VFR pairs constitute one logical value. Three integers. That is the complete type system.

A data stream is a sequence of batches terminated by a zero header:

[F, count, depth] [data…]

[F, count, depth] [data…]

[F, count, depth] [data…]

[0, 0, 0] end of stream

2.2 Everything Is a Batch

An entity in a game: a batch at depth 48, one pair per field. A Prolog fact: a batch where F is the predicate ID. An instruction: a VFR pair routed to the fetch unit. A pixel: a batch where V packs RGB and R holds alpha. A text character: V is the Unicode codepoint, R is the script family. An audio sample: V is the PCM amplitude. An SVDAG node: V

is the child offset, R is the child bitmask. An LLM weight: V is the trained value, R is the quantization remainder. A network packet: a batch at fixed depth with known field offsets.

Nothing in the system is not a batch. The meaning of a VFR pair comes from its position in the batch and the batch's F value. The same bit pattern is health at pair index 6, position at pair index 4, an instruction when routed to fetch, a pixel when written to the framebuffer. The bits do not carry meaning. Structure provides meaning.

2.3 Code Is Data

A program is a batch stream. Instructions are VFR pairs. The batch header describes how many instructions and their structure. The stream is stored in memory alongside entity data, fact tables, and pixel buffers.

When a batch stream is routed to a core's instruction fetch unit, it is code. When routed to the ALU as operands, it is data. The bits are identical. The destination determines the interpretation. Self-modifying programs write VFR pairs to the instruction region — ordinary integer writes to ordinary memory addresses.

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3. The Instruction Set

3.1 Thirty-Five Instructions

The complete Q-Foundational instruction set:

Arithmetic: ADD, SUB, MUL, ADDR, SUBR

Bitwise: AND, OR, XOR, NOT

Shift: SHR, SHL, SHR5, SHL5

VFR: DECOMP, RECOMP, RNORM, RACCUM, SCALE

Compare: CMP, CMPR

Load/Store: LDV, LDR, STV, STR, LDVR, STVR

Batch: BLOAD, BNEXT, BADDR

Transfer: TSEND, TRECV, TWAIT, TDONE

Control: JMP, JEQ, JLT, JGT, JDONE, HALT

All instructions are 32 bits, fixed width. 6-bit opcode supports 64 total slots, 29 reserved for future use. No floating-point instruction. No division instruction. No interrupt instruction. No exception instruction. No privilege mode instruction.

3.2 What Is Not Present

No division — division by 32 is SHR5, other division is a precomputed reciprocal multiply. No floating-point — all arithmetic is integer. No branch prediction — conditional jumps evaluate flags set by CMP, resolved in-order. No speculative execution — no prediction means no speculation. No cache coherence — no shared memory means no coherence protocol. No virtual memory translation — each core addresses only its local physical memory. No interrupt handling — cores run to completion and set a done flag.

3.3 Universality

These 35 instructions implement:

- **Prolog evaluation:** batch filtering with CMP and conditional jumps
- **UtilityAI scoring:** fixed-point multiply chains with MUL and SHR
- **3D rendering:** SVDAG traversal with shift-mask-compare, Gouraud shading with LERP, Perlin noise with table lookup
- **Physics simulation:** position integration with ADD, collision with SUB and CMP
- **Audio mixing:** multiply-accumulate with MUL and ADD
- **Network processing:** field validation with CMP, payload extraction with LDV
- **LLM inference:** matrix multiply-accumulate with MUL and ADD
- **Text processing:** codepoint comparison with CMP, script filtering on R values

Every domain decomposes to the same primitive operations. The domains are not different activities requiring different hardware. They are different arrangements of the same transforms.

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4. The Processor

4.1 Core Architecture

Each VFR core contains:

Instruction fetch: reads instructions from local BRAM

Instruction decode: extracts opcode and operands

ALU: shift, add, subtract, multiply, bitwise, compare

Register file: V0-V15 (i32), R0-R15 (i8), batch control, flags

Local BRAM: 9 KB exclusive memory

DMA interface: explicit transfers to/from other cores

Four pipeline stages: fetch, decode, execute, writeback. In-order execution. No speculation, no reorder buffer, no reservation stations, no register renaming.

Each core occupies approximately 1,400 LUTs on FPGA fabric. One DSP48 slice provides single-cycle 32-bit multiply. 30 cores fit on a Xilinx Zynq-7020 with 88% LUT utilization.

4.2 What Is Not Present

No floating-point unit. No branch predictor. No speculative execution engine. No reorder buffer. No register renaming. No cache coherence controller. No TLB. No store buffer. No interrupt controller. No exception handler. No privilege levels.

Each absent mechanism eliminates a category of complexity. The transistors that would implement them do not exist. In their place: more cores. Where a conventional CPU fits 8-16 complex cores, a Q-Foundational die fits hundreds or thousands of minimal cores. Every transistor does useful work.

4.3 Determinism

Given the same batch input, a VFR core produces the same batch output. Always. On every implementation. Bitwise identical. This is not an engineering achievement. It is an automatic consequence of integer arithmetic with no shared mutable state. Nondeterminism is structurally impossible.

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5. The Memory Model

5.1 Strict Isolation

Each core owns a contiguous, exclusive memory range. No core can read, write, or observe any other core's memory. There is no shared address space. There is no global bus. There is no mechanism for one core to affect another's state except through explicit, permission-checked DMA transfer.

This is not an access control policy. It is a hardware fact. The address bus of each core physically connects only to its local BRAM. Accessing another core's memory is not prohibited — it is impossible. There is no wire.

5.2 Explicit Transfer

When data must move between cores, a DMA transfer is explicitly initiated. The host controller checks a permission table. If permitted, the transfer occurs. If not, nothing happens. All cross-core communication is visible, measurable, and auditable. No hidden coherence traffic. No silent bus snooping. No implicit sharing.

5.3 Security as Geometry

Buffer overflow cannot occur because all buffers are fixed-size batches with known counts and depths. Code injection cannot occur because data at entity field offsets is never routed to instruction fetch. Privilege escalation cannot occur because there are no privilege levels. Spectre-class attacks cannot occur because there is no speculation. Network packets can only write to whitelisted entity input fields at known offsets.

These are not defenses against attacks. They are absences of attack surfaces. The security of the system is a geometric property of its physical construction.

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6. The Hardware Platform

6.1 ARM+FPGA Architecture

The Xilinx Zynq-7020 System-on-Chip combines a dual-core ARM Cortex-A9 with programmable FPGA fabric on a single die.

The ARM is the brain: boot, I/O, storage, networking, display, audio, and batch dispatch. The FPGA is the muscle: 30 VFR cores processing entity batches, Prolog evaluation, UtilityAI scoring, beam-casting, and LLM inference.

ARM (Zig bare-metal): FPGA (Verilog):

Boot and init 30 VFR cores at 150 MHz

USB keyboard/mouse 9 KB BRAM per core

Gigabit Ethernet Batch dispatcher

SD card FAT32 AXI DMA engine

HDMI framebuffer Shared read-only BRAM

Audio mixing (I2S)

SQLite persistence

Scene management

Batch dispatch to FPGA

Communication: ARM writes batch parameters to memory-mapped FPGA registers, sets CONTROL=1. FPGA processes, sets STATUS=done. Results in DDR3. One register write to start. One register read to check completion.

6.2 Data Residency

Entity data loads to the FPGA once at scene start. Each frame, the ARM signals go. The FPGA runs the complete pipeline internally — all passes operate on data already in core BRAMs and registers. No data returns to DDR3 between passes. Only render-relevant fields (position, sprite, animation state — 25 bytes per entity) transfer back per frame.

6.3 Performance

100,000 entities at 60 FPS

Fused 7-pass pipeline: 165 cycles per entity

30 cores at 150 MHz

~5 watts total system power

$343 development hardware cost

6.4 Scaling

Same Verilog design, larger FPGA:

Zynq-7020: 30 cores ~100,000 entities $343

Zynq-7045: 130 cores ~350,000 entities $600

Zynq-7100: 270 cores ~700,000 entities $1,500

UltraScale+: 400 cores ~1,000,000 entities $3,000

Multi-board cluster via Ethernet. ASIC at 130nm from FPGA-validated Verilog: $50-100K for first batch of real chips. Each step validates the previous one.

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7. The Operating System

7.1 Silo OS

Zig bare-metal kernel on ARM Cortex-A9. Eight initialization stages: CPU interrupts, memory, hardware discovery, storage, peripherals, network, FPGA link, main loop.

The OS is a data loader and I/O manager. It does not schedule threads because there are none. It does not manage virtual memory because memory is statically assigned. It does not handle exceptions because there are none on the FPGA. It does not enforce security policies because security is physical.

7.2 Storage

FAT32 on SD card. SQLite on ARM for persistence. All values stored as VFR integers. No floats in the database. Scene load: SQLite query → VFR batches → DDR3 → FPGA. Scene save: FPGA → DDR3 → SQLite → SD card. The FPGA never touches storage.

7.3 Input

USB-HID keyboard and mouse processed by ARM in Zig. Raw scancodes and mouse deltas are already integers. Written to entity input fields in DDR3 at known pair offsets. The FPGA reads them as ordinary entity data.

7.4 Networking

Full TCP/IP stack on ARM in Zig. Fixed-size packets validated for exact byte count, checksum, port, source. Invalid packets silently dropped — they never existed. Valid payload bytes copied to whitelisted entity input field offsets. The FPGA never sees a network packet.

7.5 Audio

16-channel integer PCM mixing on ARM. 48 kHz, 0.35% of one ARM core. I2S DAC output via PMOD connector or HDMI embedded audio. Silo envelope system drives volume, pan, and pitch as time-bounded VFR modifiers.

7.6 Display

HDMI output from ARM processing system. Framebuffer in DDR3. FPGA writes 3D pixel buffer. ARM composites 2D UI. ARM flips double buffer. Multiple 3D viewports at different resolutions supported — each is a separate beam-cast dispatch with its own rectangle bounds.

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8. The Application Framework

8.1 Silo: Data-Only Execution

The compiled binary contains no gameplay types, no behavior code, no knowledge of what a dragon or a chair is. All behavior resides in VFR batch data: state machines, Prolog rules, UtilityAI behavior sets, logic block stacks, envelope definitions. Change the data, change the behavior. No recompilation. No restart.

8.2 The Entity

Every object is the same struct. 48 VFR pairs. Same layout for every entity. The difference between a dragon and a chair is which fields contain data, not what type they are. Field replacement allows semantic relabeling: health becomes bank balance, stamina becomes credit limit. Same i32, different meaning determined by a replacement table the UI reads.

8.3 The Execution Funnel

Each frame, each entity passes through:

State Machine → Prolog → UtilityAI → Logic Blocks → Envelopes

Each layer filters scope. State machines determine what the entity is doing overall. Prolog checks preconditions. UtilityAI scores candidate behaviors. Logic blocks execute the winner. Envelopes apply time-bounded modifications. Each layer is a batch operation on the same entity data.

8.4 Prolog as Batch Filtering

Facts are batches where F is the predicate ID. Rule evaluation is chained batch filters with integer comparison. Variable binding is reading a column from a matched row. No strings. No unification algorithm. No backtracking search.

8.5 Scene Isolation

Scenes map to cores or groups of cores. Each scene's entity data lives in its own memory region. Cross-scene access requires explicit DMA with permission checking. Default deny. This mirrors the hardware memory isolation at the application level.

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9. The Rendering Pipeline

9.1 Sovereign Volumetric Engine

No GPU. No triangles. No textures. No OpenGL. No Vulkan.

The world is a database of Build Boxes. Each box contains an SVDAG — a Sparse Voxel Directed Acyclic Graph. Nodes are VFR pairs: V is the child offset, R is the 8-bit child bitmask. Bit 1 means solid. Bit 0 means air.

9.2 Beam-Casting

For each pixel, a beam drills the SVDAG. At each depth level: compute octant (shift), check bitmask (AND), count preceding bits (popcount), index to child (ADD). First solid leaf terminates the beam. Zero overdraw by construction.

9.3 The 3×3 Block Pattern

The screen tiles into 3×3 blocks. The center pixel is a scout beam — full drill. The 8 neighbors verify the scout's path and reshade if matching, full drill only if divergent. 4x speedup on average. Adjacent blocks share path prefixes for additional 2.7x from coherence.

9.4 Shading

Two layers, both integer. Gouraud: trilinear interpolation across 8 box corner colors. 7 LERPs, 28 ALU operations. Smooth artist-controlled gradients. Perlin: permutation table lookup and integer gradient interpolation. 50 ALU operations. Infinite non-repeating surface detail.

9.5 Procedural Hit Evaluator

After SVDAG reports solid, the PHE optionally rejects the hit based on math — edge softening, implicit shapes (cylinders, spheres), noise-driven erosion. Solid can become air, never the reverse. Curves and organic shapes without changing geometry storage.

9.6 Upscaling

Internal resolution lower than output. Edge-aware upscale uses scout metadata (box ID, material, local coordinates) to preserve silhouettes while smoothing interiors. Five tiers from nearest neighbor to geometry-aware reshade. Hardware tier determines internal resolution; upscale quality scales the visual output.

Zynq-7020: 640×360 internal $\rightarrow$ 1280×720 output 5.2 ms render

Zynq-7020: 960×540 internal $\rightarrow$ 1920×1080 output 8.3 ms render

Zynq-7045: 960×540 internal $\rightarrow$ 2560×1440 output 2.6 ms render

Zynq-7100: 1920×1080 internal $\rightarrow$ 3840×2160 output 2.2 ms render

9.7 Geometric Deduplication

Multiple Build Boxes reference the same SVDAG root. 1000 pillars share one node set. Visual variety from per-box mutator settings — noise rotation, scale, warp. A few bytes of metadata per box. Zero additional geometry memory. SVDAG nodes cached in BRAM serve all references.

9.8 Automatic LOD

Drill depth determines resolution. Distant geometry drills fewer levels — 50 cycles instead of 200. No separate LOD meshes. No transitions. Same SVDAG, different traversal depth.

9.9 2D UI

Clay layout library on ARM. Outputs rectangles, text positions, colors, borders. ARM draws directly to framebuffer: filled rectangles as pixel writes, text as bitmap font blits, alpha blending as integer LERP per channel. No GPU. Approximately 2 ms for a full UI pass.

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10. Inline Intelligence

10.1 The LLM Is a Batch Pass

The language model is not a service. It is not an API call. It is a batch operation in the frame pipeline. When an entity encounters a situation no rule covers, the LLM generates candidate rules as VFR integer tuples. Prolog verifies them. Verified rules enter the active rule set. Rejected rules are discarded.

10.2 Neural Inference as Integer Math

The LLM forward pass is matrix multiply-accumulate on integer weight batches. MUL, ADD, SHR. Same instructions as everything else. A 100M parameter model generates

one rule in approximately 3 frames on the Zynq-7020. The entity continues existing behavior during generation, then seamlessly transitions to the new behavior.

10.3 No Hallucination by Construction

The LLM outputs integer tuples describing candidate rules. Prolog verification checks every predicate ID against the knowledge base, every argument against valid ranges, every action against the entity's valid action set. Invalid tuples are discarded before they can affect anything. There is no path from LLM output to entity behavior that bypasses verification.

10.4 The Knowledge Base

The KB is a set of VFR fact batches in DDR3. Entity state IS part of the KB — facts generated from entity fields every frame. Persistent facts accumulate from game events. Designer rules load at scene start. LLM-generated rules appear during play. All are queried by the same Prolog filter.

The KB has no size limit, no positional degradation, no session boundary, no information loss from compaction. Turn 10,000 is as coherent as turn 1. Cold facts evict to SD card. Nothing is deleted.

10.5 Triveritas Evaluation

Every claim is evaluated on three dimensions: logical validity (contradictions), mathematical coherence (type and range checking), empirical anchoring (provenance depth). Three batch filter passes. Integer comparison and counting. The classification is a 3-bit bitmask.

10.6 Domain Eating

Adding a knowledge domain: write a parser that outputs VFR fact batches with provenance. Load batches into KB. Immediately queryable. No neural network retraining. No weight updates. The FPGA doesn't know it switched from game AI to medical knowledge. It's filtering integer batches.

10.7 VFR Text

Characters are VFR pairs: V is the Unicode codepoint, R is the script family. Fixed 5 bytes per character. O(1) indexing. Instant script detection from one byte. The LLM's Term tokenizer produces VFR term batches with provenance at each depth level. Every token carries its grammatical role, source file, byte offset, and version.

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11. The Error Model

11.1 Errors That Cannot Occur

Divide by zero: no divide instruction. NaN: no floating point. Null pointer: no pointers on FPGA. Buffer overflow: all batches are fixed-size. Stack overflow: no stack, no recursion. Type mismatch: one type. Race condition: no shared memory. Deadlock: no locks.

These are not handled errors. They are errors the architecture makes structurally impossible.

11.2 Errors That Can Occur

DMA hardware failure: retry forever. Core halt on undefined opcode: compiler bug. SD card failure: machine cannot load, halts. Network corruption: dropped silently. USB disconnection: input fields stop updating, game continues, device reconnects.

The system either works or the hardware is broken. The concept of a software error effectively ceases to exist.

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12. The Philosophy

12.1 The Collapse of Distinctions

Modern computing maintains separations: code versus data, types versus values, hardware versus software, security versus architecture, intelligence versus arithmetic. Each separation creates a boundary, a protocol, complexity, failure modes, and defensive machinery.

The Q-Foundational Stack dissolves these separations. Code is a batch routed to fetch. Data is a batch routed to the ALU. A type is an F value and a depth. Security is the absence of wires between cores. Intelligence is a particular arrangement of batch filters. The separations were conventions, not necessities.

12.2 Computation as Arrangement

The CPU does not run a game. It resolves the next arrangement of integers from the current arrangement according to 35 rules. Like a cellular automaton. Like crystallization. Like physics itself, which does not execute laws — it is laws. Integers in one configuration become integers in another configuration. There is no execution. There is only transformation.

12.3 The Thesis

Data is data. Computation is a type of data. Values are a type of data. Code is a type of data. Types are a type of data. Intelligence is a type of data. Security is a property of geometry. Scale is a number in a header. The universe of computing reduces to `[V: i32, R: i8]` at some F and depth.

All the way down.

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13. The Complete Specification Suite

Document Defines

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1. Integer ALU Based Computing VFR representation, architectural case
2. ISA Specification v1.0 35 instructions, encoding, opcode map
3. Compiler Specification v1.0 VFR-Lang, batch emission, fusion
4. Silo-VFR Mapping v1.0 Game engine on VFR hardware
5. Motherboard Specification v2.0 Zynq ARM+FPGA platform, performance
6. Rendering Pipeline v2.0 Beam-casting, shading, upscaling
7. FPGA Implementation v1.0 Verilog design, build plan, timeline
8. System Integration v1.0 I/O, storage, networking, audio, errors
9. The Philosophy of Q-Computation Why this works
10. LLM-Prolog Inline Intelligence v1.0 Neural-symbolic as batch operations
11. The Q-Foundational Stack (this document) The complete architecture

13.1 Build Order

Phase Weeks Deliverable

ALU + sim 1-2 vfr_alu.v verified in simulation

Single core 3-4 vfr_core.v verified in simulation

Hardware 5-6 One core on FPGA, ARM C test program

Multi-core 7-8 30 cores, batch dispatcher working

Zig port 9-10 ARM running Zig bare-metal kernel

Silo engine 11-12 Entity processing on FPGA, verified against software

Rendering 13-16 Beam-casting, pixels on HDMI display

Integration 17-20 Complete system at 60 FPS

13.2 Hardware Cost

Digilent Zybo Z7-20 $280

MicroSD card $10

USB keyboard $15

USB mouse $10

HDMI cable $8

Power supply $15

USB debug cable $5

PMOD I2S DAC $10

Total: $353

All development tools free: Vivado WebPACK, Zig compiler, Verilator, GTKWave.

13.3 Performance Summary

Metric	Value
Entities at 60 FPS	~100,000
Entity pipeline cycles	165 per entity (fused 7-pass)
Render internal resolution	960 × 540 (upscaled to 1080p)
Render time	8.3 ms per frame
VFR cores	30 at 150 MHz
Total system power	~5 watts
LLM inference (100M params)	6.67 ms per token
Development hardware cost	$343
Total Verilog	~1,650 lines
Total testbench	~1,000 lines
Instructions in ISA	35

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14. The New World

A world where a $343 board runs 100,000 game entities with AI, physics, volumetric 3D rendering, networking, and audio at 60 FPS on 5 watts.

A world where security is a physical property of the hardware, not a software layer that can be bypassed.

A world where every computation is deterministic. Run it twice, get the same answer. Run it on a different machine, get the same answer. Debug by replaying the exact input that produced the bug.

A world where the game engine, the operating system, the renderer, the AI, the database, and the network stack all use the same data format, the same instructions, and the same processing model.

A world where adding intelligence to a game entity is the same operation as adding a new behavior rule — write integers to a batch. Where the AI cannot hallucinate because verification sits between generation and use.

A world where adding a knowledge domain — medicine, law, engineering, any field — means writing a parser that outputs integer batches. No retraining. No fine-tuning. Load and query.

A world where text in any language is the same 5-byte fixed-width representation as every other value. Where script detection is one byte. Where cross-language search is integer comparison.

A world where the complexity of modern computing — the branch predictors, the speculation engines, the coherence protocols, the floating-point units, the exception handlers, the virtual memory translators, the privilege levels, the security layers — is recognized as the accumulated consequence of two decisions made forty years ago: approximate numbers with floats, and share memory between cores.

Different decisions. Different consequences. Different world.

One atom: `[V: i32, R: i8]`.

One container: the batch.

One instruction set: 35 operations.

One memory model: isolation.

One processing model: transform batches.

One philosophy: data is data, and everything is data.

The Q-Foundational Stack.

Registry of constituent specifications: ISA v1.0, Compiler v1.0, Silo-VFR Mapping v1.0, Motherboard v2.0, Rendering Pipeline v2.0, FPGA Implementation v1.0, System Integration v1.0, Philosophy of Q-Computation, LLM-Prolog Inline Intelligence v1.0

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C MATH-0-2026

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-10-2026**] Grand Unification. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026

[**CKS-MATH-104-2026**] Grand Unification v23. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS MATH-104-2026

[**CKS-LEX-12-2026**] Measurement Systems in the $\mathbb{Q}$-Substrate. Github: https://github.com/ghowland/cks/tree/main/paper LEX-12-2026